Progress report: Tense and Aspect in Matrix customization

Laurie Poulson
Matrix Developers Group
University of Washington, Seattle

Outline

- Goals
- Phenomena
- Details
- Results
- Issues
- Future plans

Goals

- Provide a useful jump start for tense and aspect implementations
  - Stay agnostic (mostly) regarding analyses
  - Make the questionnaire flexible
- Implement in stages
- Provide foundation for later stages

Phenomena

- Current working version:
  - Grammatical tense
  - Grammatical (viewpoint) aspect
  - Implements common forms:
    - Concatenative inflection
    - Basic auxiliary verb constructions (in simple, indicative, main clauses only)
- Planned: (I'll come back to that)
- Not planned: Mood, Voice, serial verb constructions, dependent clauses, etc.

Assumptions

- Tense and aspect are features of an event
- Tense and aspect values may be:
  - Lexically specified
  - Introduced through inflectional rules
  - Underspecified until constrained
  - Other ??
- Auxiliaries function as a head and:
  - May or may not introduce a relation (with event)
  - May place constraints on their complements
  - Languages with auxiliaries distinguish between finite and non-finite verb forms

Overview: Questionnaire

- Define tense and aspect feature hierarchies
  - Tense:
    - Choose elements of a common hierarchy
    - Build your own hierarchy
  - Aspect: Build your own hierarchy
- If applicable:
  - Define auxiliaries and constraints on their complements
    - e.g., constrain FORM on the complement verb
  - Associate feature values with verbal inflection
    - e.g., constrain PNG values on the verb's subject
    - e.g., specify value of tense on the verb

Demo questionnaire

questionnaire

English-like TDL excerpts:

```
;;; Tense
non-future := tense.
past := non-future.
present := non-future & non-past.
non-past := tense.
future := non-past.

;;; Aspect
imperfective := aspect.
perfective := aspect.
progressive := imperfective.
habitual := imperfective.
```

English-like TDL excerpts:

```
subj-raise-aux-no-pred := subj-raise-aux & raise-sem-lex-item &
    [ ARG-ST < [  ],
                [ LOCAL.CAT.HEAD [ AUX -,
                                   FORM presentparticiple ] ] > ].
aux-aux-lex := subj-raise-aux-no-pred & tenseaspect-rule-dtr &
    [ INFLECTED - ].

subj-raise-aux-with-pred := subj-raise-aux & norm-sem-lex-item &
    trans-first-arg-raising-lex-item-1 &
    [ ARG-ST < [  ],
                [ LOCAL.CAT.HEAD.FORM infinitive ] > ].
modal-aux-lex := subj-raise-aux-with-pred &
    [ INFLECTED - ].
```

English-like TDL excerpts:

```
present3sg-lex-rule := infl-ltow-rule & tenseaspect-lex-rule &
[ SYNSEM.LOCAL [ CONT.HOOK.INDEX.E.TENSE present,
                 CAT [ HEAD.FORM finite,
                       VAL.SUBJ.FIRST.LOCAL.CONT.HOOK.INDEX.PNG [ NUM singular,
                                                                  PER 3rd ] ] ] ].

past-lex-rule := infl-ltow-rule & tenseaspect-lex-rule &
[ SYNSEM.LOCAL [ CONT.HOOK.INDEX.E.TENSE past,
                 CAT.HEAD.FORM finite ] ].

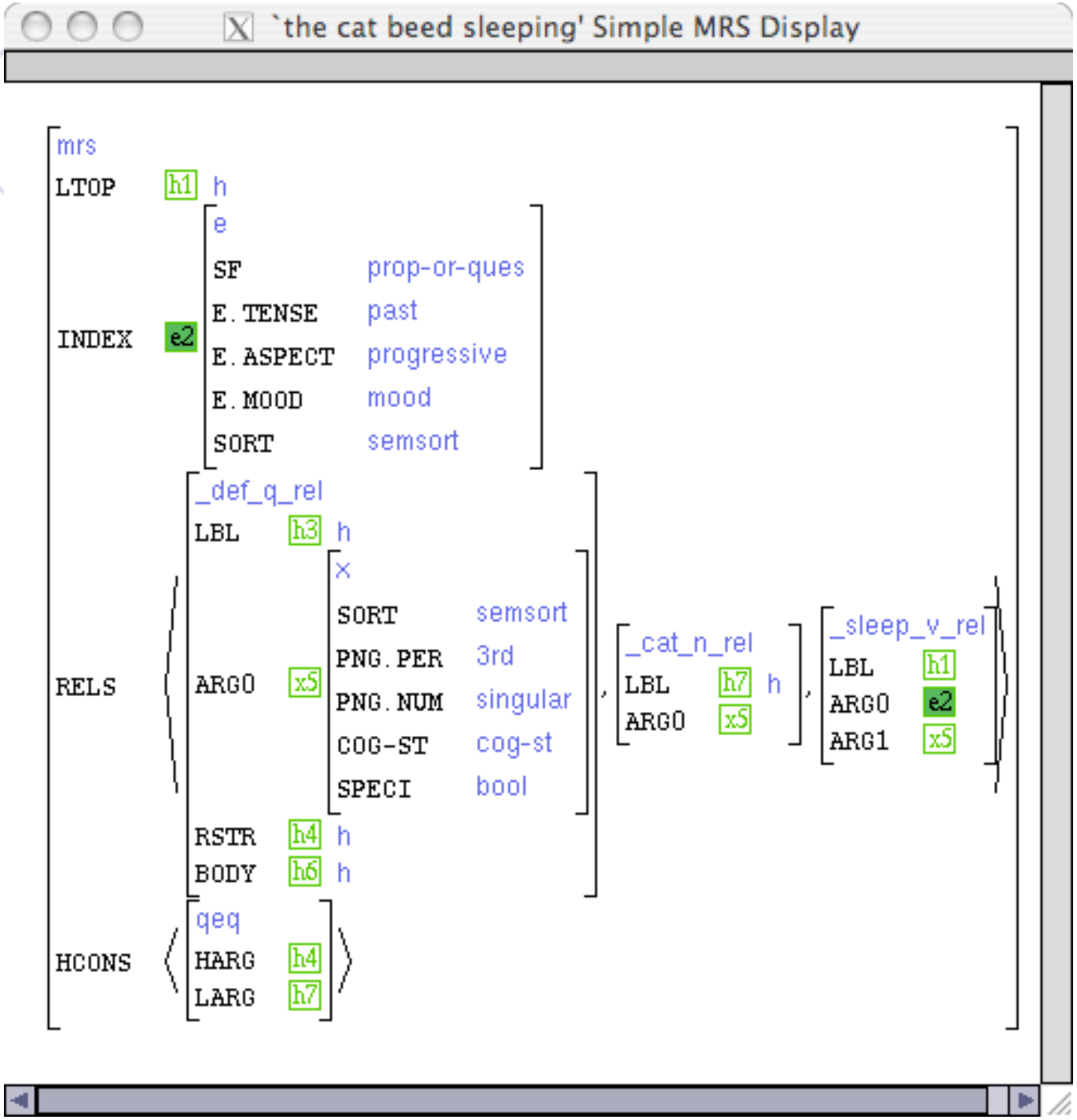
prespart-lex-rule := infl-ltow-rule & tenseaspect-lex-rule &
[ SYNSEM.LOCAL [ CONT.HOOK.INDEX.E.ASPECT progressive,
                 CAT.HEAD.FORM presentparticiple ] ].
```

Sample English-like parse results:

1	the cat sleeps	1 15	MRS: present, 3rd-sg
2	*the cat sleep	0 14	
3	the cats sleep	1 17	MRS: present, 3rd-pl
4	*the cats sleeps	0 16	
5	*sleep the cat	0 13	
6	*sleeps the cat	0 14	
7	the cat sleeped	1 13	MRS: past, 3rd-sg
8	the cats sleeped	1 15	MRS: past, 3rd-pl
9	the cat bes sleeping	1 20	MRS: present, progressive, 3rd-sg
10	*bes sleeping the cat	0 19	
11	the cat beed sleeping	1 18	MRS: past, progressive, 3rd-sg
12	*the cat beed sleep	0 17	
13	*the cat beed sleeped	0 16	
14	the cat can sleep	1 19	MRS: present (can), 3rd-sg
15	*the cat cans sleep	0 18	
16	*the cat caned sleep	0 16	
17	*the cat can sleeping	0 16	
18	*the cat can sleeped	0 16	
19	the cat can be sleeping	1 25	MRS: present(can), 3rd-sg, progressive(sleep)

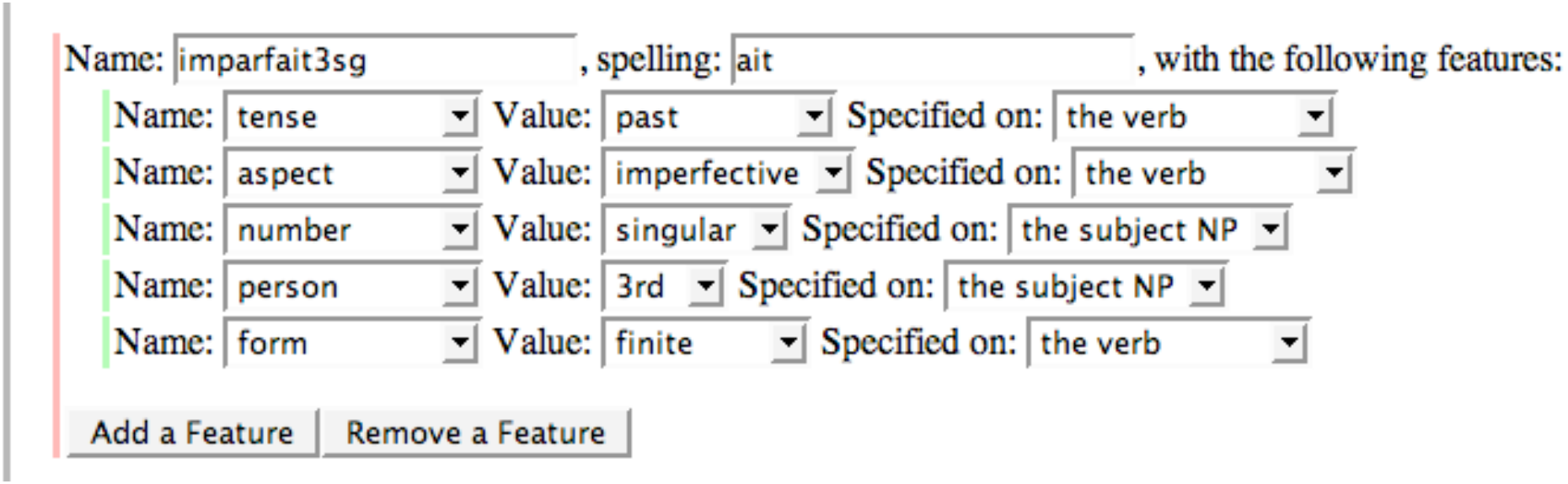
Example

MRS:


`the cat beed sleeping' Simple MRS Display
mrs
LTOP h1 h
INDEX e2
e
SF prop-or-ques
E.TENSE past
E.ASPECT progressive
E.MOOD mood
SORT semsort
RELS
_def_q_rel
LBL h3 h
ARG0 x5
x
SORT semsort
PNG.PER 3rd
PNG.NUM singular
COG-ST cog-st
SPECI bool
RSTR h4 h
BODY h6 h
_cat_n_rel
LBL h7 h
ARG0 x5
_sleep_v_rel
LBL h1
ARG0 e2
ARG1 x5
HCONS
qeq
HARG h4
LARG h7

French-like:

morpheme definition:

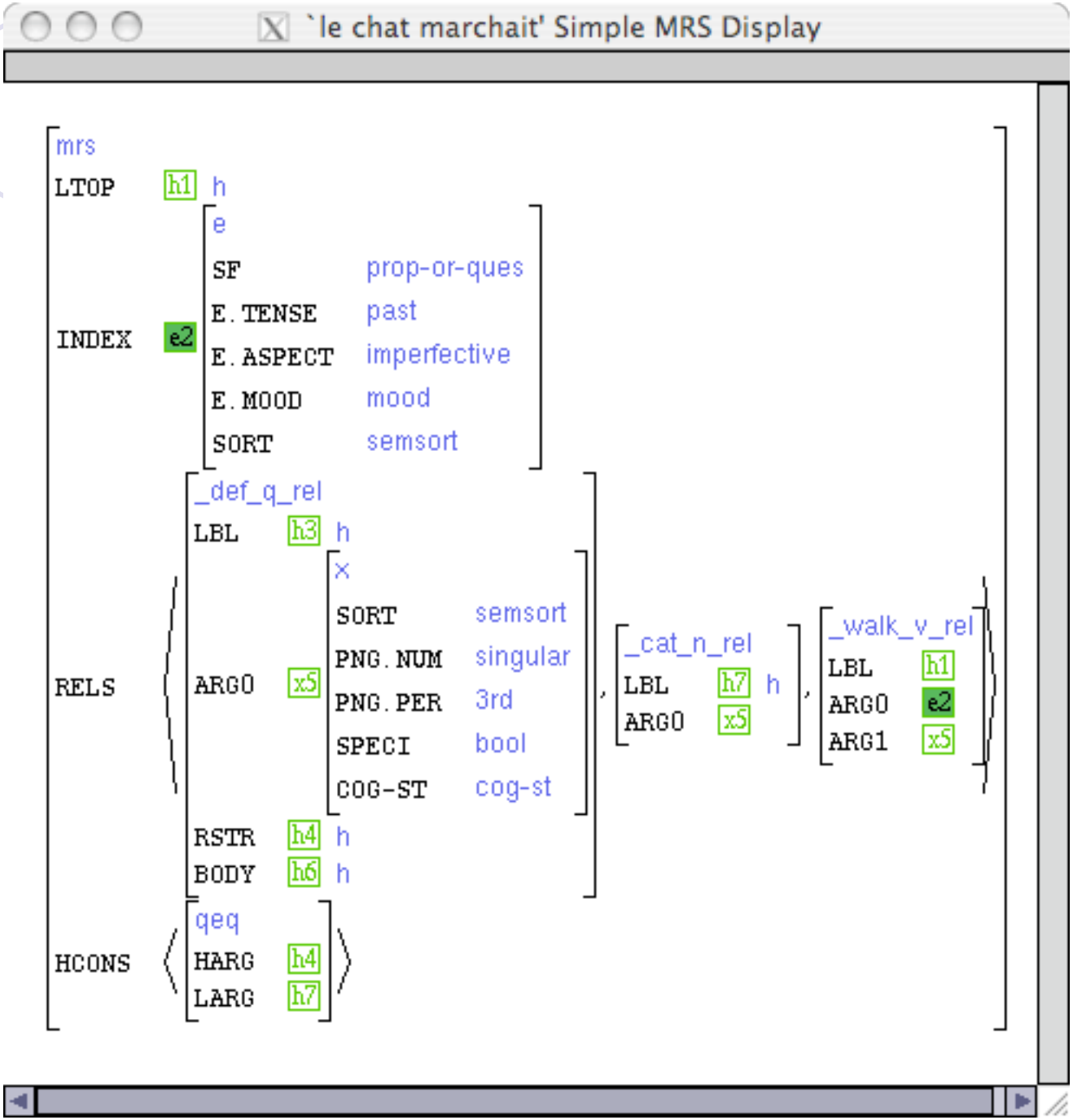


resulting inflectional rule type:

```
imparfait3sg-lex-rule := verbcong-lex-rule &
[ SYNSEM.LOCAL [ CONT.HOOK.INDEX.E [ TENSE past,
                                     ASPECT imperfective ],
                 CAT [ VAL.SUBJ.FIRST.LOCAL.CONT.HOOK.INDEX.PNG [ NUM singular,
                                                                  PER 3rd ],
                       HEAD.FORM finite ] ] ].
```

Example

MRS:


`le chat marchait' Simple MRS Display
mrs
LTOP h1 h
INDEX e2
e
SF prop-or-ques
E.TENSE past
E.ASPECT imperfective
E.MOOD mood
SORT semsort
RELS
_def_q_rel
LBL h3 h
ARG0 x5
x
SORT semsort
PNG.NUM singular
PNG.PER 3rd
SPECI bool
COG-ST cog-st
RSTR h4 h
BODY h6 h
_cat_n_rel
LBL h7 h
ARG0 x5
_walk_v_rel
LBL h1
ARG0 e2
ARG1 x5
HCONS
qeq
HARG h4
LARG h7

Future plans: Imminent

- Constraints based on verb classes needed
  - Russian: “perfective” (vs. “imperfective”) verbs can take a secondary imperfective affix
  - French:
    - Avoir vs être verbs
    - er, ir, re verbs
  - English: Statives are not progressivized
- Plan:
  - Define verb type hierarchies
  - Distinguish types with feature values
  - Expand the questionnaire to elicit aux-complement verb classes as appropriate

Future plans: Imminent (cont.)

- Particles
  - Plan: Implement as modifiers
- Nominal tense (propositional)
  - Plan:
    - Define (possibly homophonous) tense features on nouns
    - Constrain as agreement

Future plans: Farther out

- Later:
  - Perfect
    - issues: Multiple events: state + anterior event (?), Temporal value of un-tensed events
  - Interaction with adverbs
    - issues: tense vs. time, scope
  - Situation aspect
    - issues: Compositionality, Constraints
- Way out:
  - ??? Interaction with discourse